# Adaptation Strategy in Mission Bay

*Supervisor：Theuns Henning*

*Haochen Zhang*

## 1 STUDY AREA

The study area comprises the Mission Bay shoreline and back-beach neighbourhood along Tamaki Drive, from Bastion Point to the eastern bend of Tamaki Drive (~1.45 km of coast). The seaward boundary follows the mapped coastal-inundation envelope. The landward boundary follows the landward extent of Auckland Council GeoMaps' coastal-inundation layer for 18.1% AEP with +2.0 m sea-level rise. Topographically, the area forms a saddle, with low flats flanked by higher ground to the west and east.


Figure 1. Study Area - Mission Bay

## 2 ASSETS CONSIDERED WITHIN THE AREA

- Tamaki Drive coastal section
- Hard coastal structures
- Residential blocks
- Local street network in the back-beach blocks
- Mission Bay beach and foredune
- Selwyn Reserve
- Tagalad Reserve North
- Bastion Point Recreation Reserve

## 3 HAZARD TYPES

### 3.1 *Coastal Inundation*

Short-term marine flooding of the shore belt and back-beach flats driven by tides, storm surge and sea-level rise (SLR). Two map families are used:

a. Mean High Water Spring Tides (MHWS) + SLR (0.5 - 2.0)
b. Storm-tide AEP (18.1%, 4.9%, 2%, 1%) + SLR (0.5 - 2.0)

### 3.2 *Coastal Erosion*

Progressive landward retreat of the active beach edge under waves, storms and SLR. The assessment uses ASCIE Level A (Regional) lines for 2050, 2080, 2130 (RCP8.5 / 8.5+).

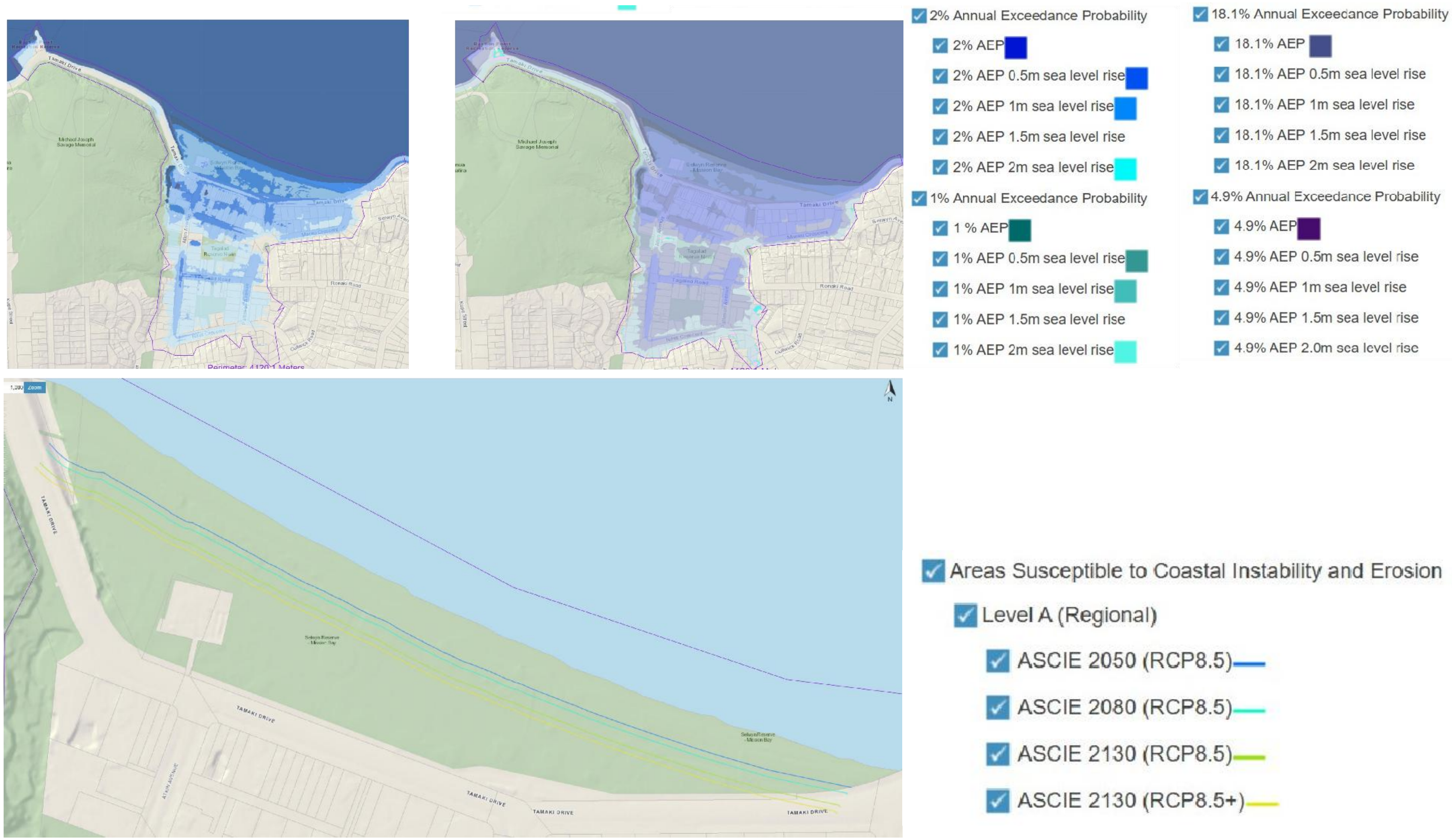


Figure 2. Hazard Type.

# 4 RESULT OF RISK ANALYSIS

| Type | Year/SLR | Storm | Tamaki Drive (Exposure) | Local Streets (Exposure) | Hard Structures (Exposure) | Residential Blocks (Exposure) | Beach & Foredune (Exposure) | Selwyn Reserve (Exposure) | Tagalad Reserve North (Exposure) | Bastion Point Recreation Reserve (Exposure) | Aggregated Exposure | Aggregated Vulnerability | Aggregated Consequence |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| MHWS | 0.5M | - | L | L | M | L | H | L | L | L | L | L | Insignificant |
| MHWS | 1.0M | - | M | M | H | L | H | M | L | L | M | M | Moderate |
| MHWS | 1.5M | - | H | H | H | M | H | H | L | L | H | H | Major |
| MHWS | 2.0M | - | H | H | H | H | H | H | H | M | H | H | Major |
| Inundation | 0M | 18.1% AEP | L | L | M | L | H | L | L | L | L | L | Insignificant |
| Inundation | 0.5M | 18.1% AEP | L | L | M | L | H | M | L | L | L | L | Insignificant |
| Inundation | 1.0M | 18.1% AEP | M | M | H | M | H | H | L | L | H | M | Major |
| Inundation | 1.5M | 18.1% AEP | H | H | H | H | H | H | H | M | H | H | Major |
| Inundation | 2.0M | 18.1% AEP | H | H | H | H | H | H | H | H | H | H | Major |
| Inundation | 0M | 4.9% AEP | L | L | M | L | H | L | L | L | L | L | Insignificant |
| Inundation | 0.5M | 4.9% AEP | L | L | M | L | H | M | L | L | L | L | Insignificant |
| Inundation | 1.0M | 4.9% AEP | M | M | H | M | H | H | L | L | H | M | Major |
| Inundation | 1.5M | 4.9% AEP | H | H | H | H | H | H | H | M | H | H | Major |
| Inundation | 2.0M | 4.9% AEP | H | H | H | H | H | H | H | H | H | H | Major |
| Inundation | 0M | 2% AEP | L | L | M | L | H | L | L | L | L | L | Insignificant |
| Inundation | 0.5M | 2% AEP | L | L | M | L | H | M | L | L | L | L | Insignificant |
| Inundation | 1.0M | 2% AEP | M | M | H | M | H | H | L | L | H | M | Major |
| Inundation | 1.5M | 2% AEP | H | H | H | H | H | H | H | M | H | H | Major |
| Inundation | 2.0M | 2% AEP | H | H | H | H | H | H | H | H | H | H | Major |
| Inundation | 0M | 1% AEP | L | L | M | L | H | L | L | L | L | L | Insignificant |
| Inundation | 0.5M | 1% AEP | L | L | M | L | H | M | L | L | L | L | Insignificant |
| Inundation | 1.0M | 1% AEP | M | M | H | M | H | H | L | L | H | M | Major |
| Inundation | 1.5M | 1% AEP | H | H | H | H | H | H | H | M | H | H | Major |
| Inundation | 2.0M | 1% AEP | H | H | H | H | H | H | H | H | H | H | Major |
| Erosion | ASCIE 2050 (RCP8.5) | - | L | L | M | L | H | M | L | L | L | L | Insignificant |
| Erosion | ASCIE 2080 (RCP8.5) | - | M | L | M | L | H | H | L | L | M | L | Minor |
| Erosion | ASCIE 2130 (RCP8.5) | - | M | L | H | L | H | H | L | L | H | M | Major |
| Erosion | ASCIE 2130 (RCP8.5+) | - | M | L | H | L | H | H | L | L | H | M | Major |

# 5 PATHWAY ADAPTATION

## 5.1 *Unified Baseline Adaptation Stage (0m - +1.0m SLR)*

In the initial stages of sea-level rise (0m to +1.0m), a unified site-wide strategy of maintenance and incremental defence is adopted. Up to +0.5m, the focus is on fixing backflow risk points, minor raising of crests and kerbs, lifting critical path nodes, and commencing dune restoration with annual monitoring.

As SLR approaches +1.0m, actions escalate to incremental topping-up and sealing of weak seawall segments, establishing floodable lawns and linear storage at the reserves, and providing self-protection guidance for first-row properties while adjusting maintenance standards for more frequent inspections.

## 5.2 *Lifeline Protection (> +1.0m SLR)*

Beyond +1.0m SLR, Tamaki Drive requires robust protection as lifeline infrastructure. Between +1.0m and +1.5m, actions include re-grading the road to raise sags, preparing rapid-deployment barrier nodes, adding toe dissipation and anti-scour works, and raising or duplicating critical utilities as required.

Beyond +1.5m, the strategy shifts to transformative engineering, aiming to establish a continuous elevation and water-tightness line along the entire corridor.

## 5.3 *Public Space Accommodation (> +1.0m SLR)*

Reserves and beaches adopt an 'Accommodate' pathway, living with water. Key actions include ongoing beach nourishment to reach design volumes and transitioning areas like Selwyn Reserve from purely recreational spaces into zones with active flood management functions, accepting their role as temporary buffers during storms.

## 5.4 *Residential Managed Retreat (> +1.0m SLR)*

Residential areas transition gradually from 'Accommodate' to 'Managed Retreat' starting at +1.0m SLR. The initial focus (+1.0m to +1.5m) is on enabling ground-floor use conversion to non-residential purposes and adopting floodable building designs.

Beyond +1.5m, or if annual inundation thresholds are persistently exceeded, a full retreat strategy is triggered: securing reserved retreat corridors, utilizing planning tools to limit new risks, and commissioning feasibility and funding studies for selective relocation.

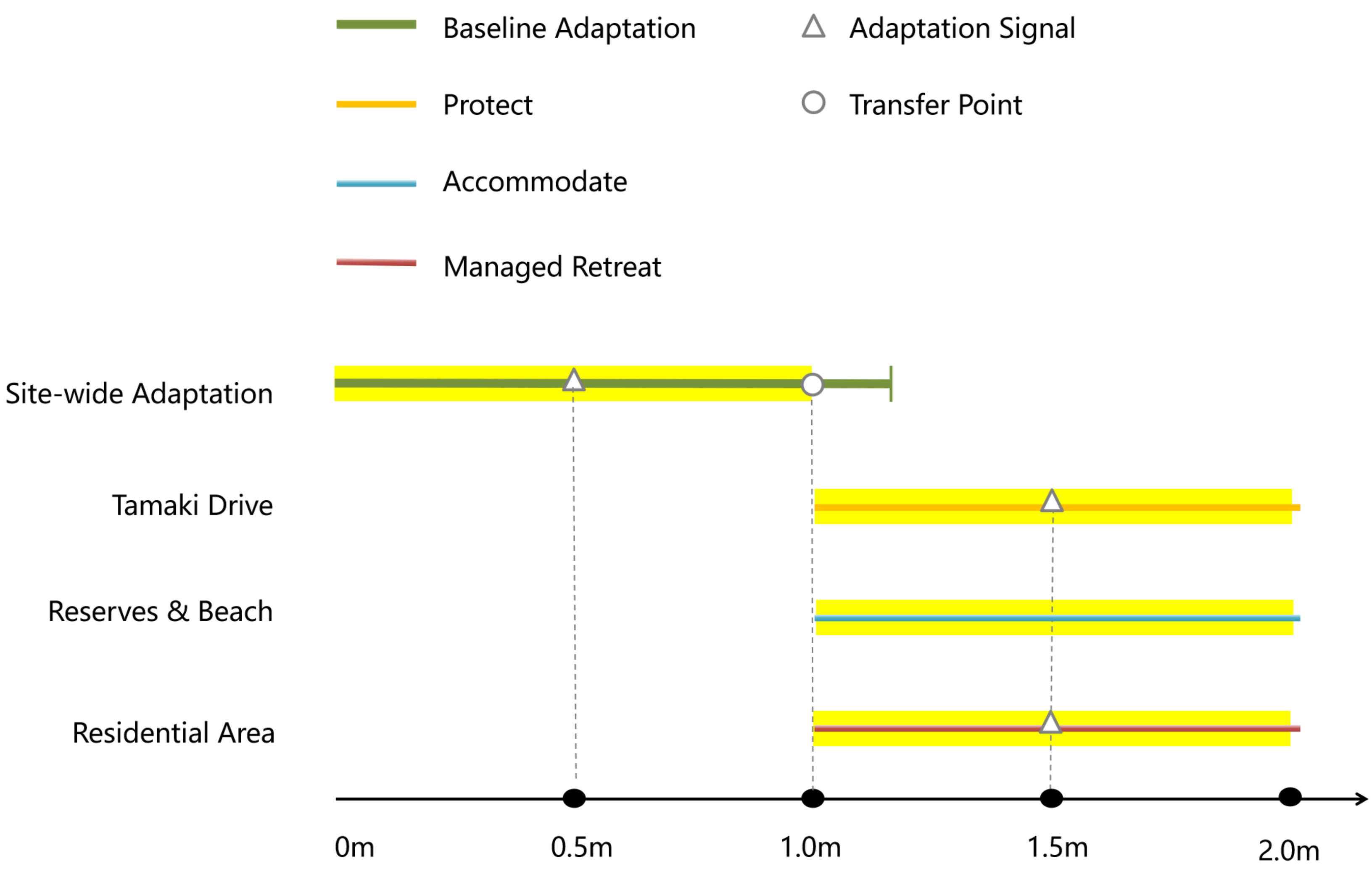


Figure 3. Adaptation Pathway.